

University Important questions

Q1. Measurement and chronology of permanent maxillary Central incisor.

Q2. Measurement and chronology of permanent maxillary lateral incisor

Q3. Difference between maxillary Central and lateral incisor.

Q4. Measurement and chronology of permanent mandibular Central incisor.

Q5. Measurement and chronology of permanent mandible lateral incisor.

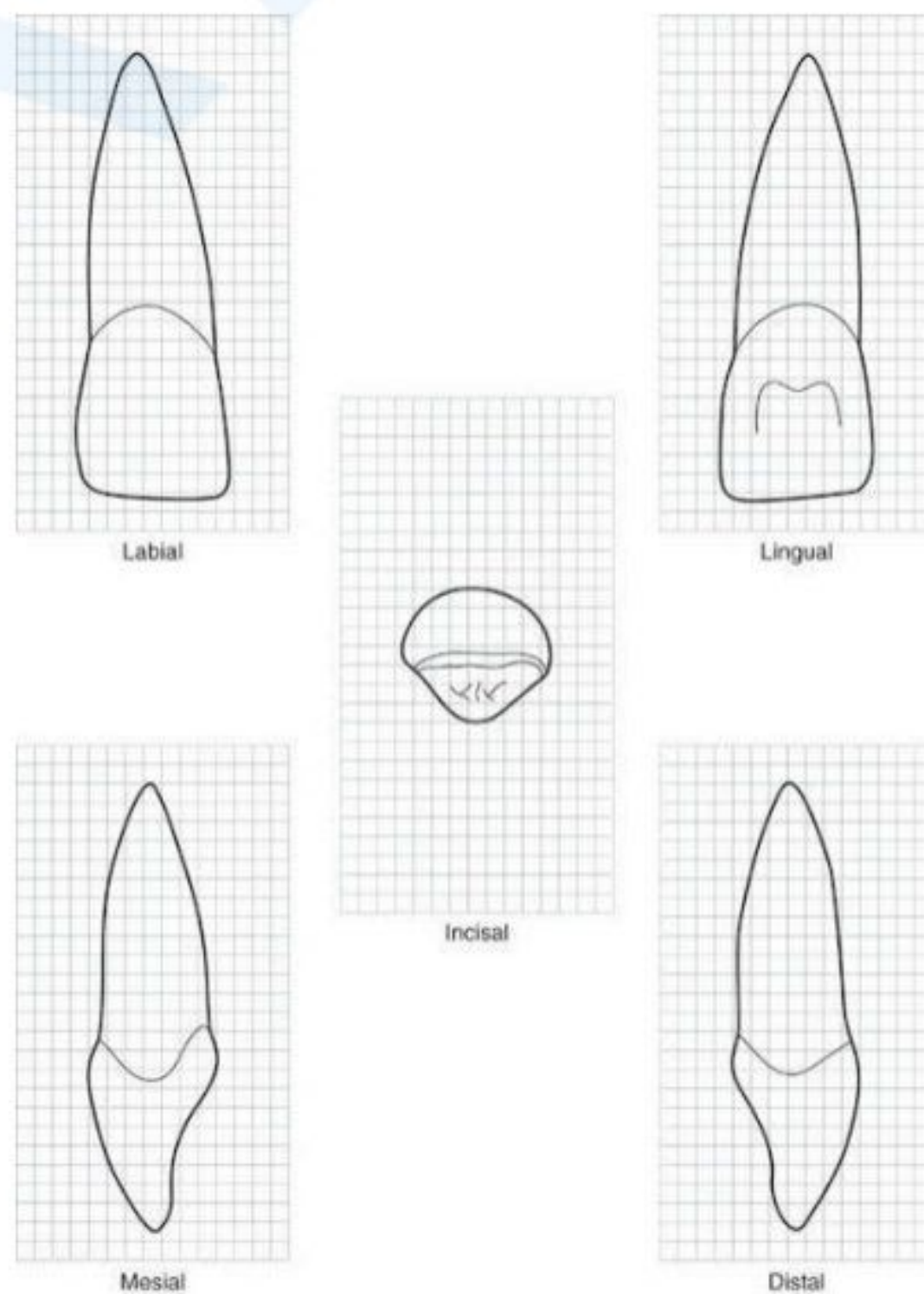
Q6. Difference between mandibular permanent Central and lateral incisor.

Q7. Line angles and point angles in maxillary Central incisor.

The most important questions are:

Q1, Q2, Q3, Q4, Q5, Q6, Q7.

Q1. Measurement and chronology of permanent maxillary Central incisor.



Permanent Dentition

- **Completion:** Occurs between **18 to 25 years** of age if the **third molar** is included.
- **Central Incisor:** Second permanent tooth to emerge, with emergence occurring between **6 and 7 years**.

Chronology of Permanent Maxillary Central Incisor

Measurement/Event	Age (Months/Years)
First Evidence of Calcification	3-4 months
Crown Completion	7-8 years
Germination	10 years
Eruption	4-5 years

General Features of Maxillary Permanent Central Incisor

- **Largest** of all incisors.
- **Two in number**, one on either side of the midline.
- Most prominently seen in the oral cavity.
- **Occludes** with lower central and lateral incisors.

Labial Surface

- Convex both **mesiodistally** and **cervicoincisally**.

- Incisal third is **flattened** with two faint **developmental grooves** dividing it into **three mamelons**.
- **Mesial and distal borders** taper cervically.
- Distoincisal angle is **rounded**, mesioincisal angle is **sharper**.
- Cervical line is **convex rootwards**.
- Maximum **mesiodistal width** is at the middle-third of the crown.

Palatal Surface

- **Concave** with a pronounced convex **cingulum**.
- **Prominent mesial and distal marginal ridges** run to the incisal edge enclosing the **lingual fossa**.
- Overall surface area is less than the **labial surface** due to **lingual convergence** of proximal surfaces.
- Cervical line is **convex rootwards**.

Mesial and Distal Surfaces

- When viewed from the incisal edge, the crown **obscures the root**.
- Labial side appears **broad and flat** in the incisal third.

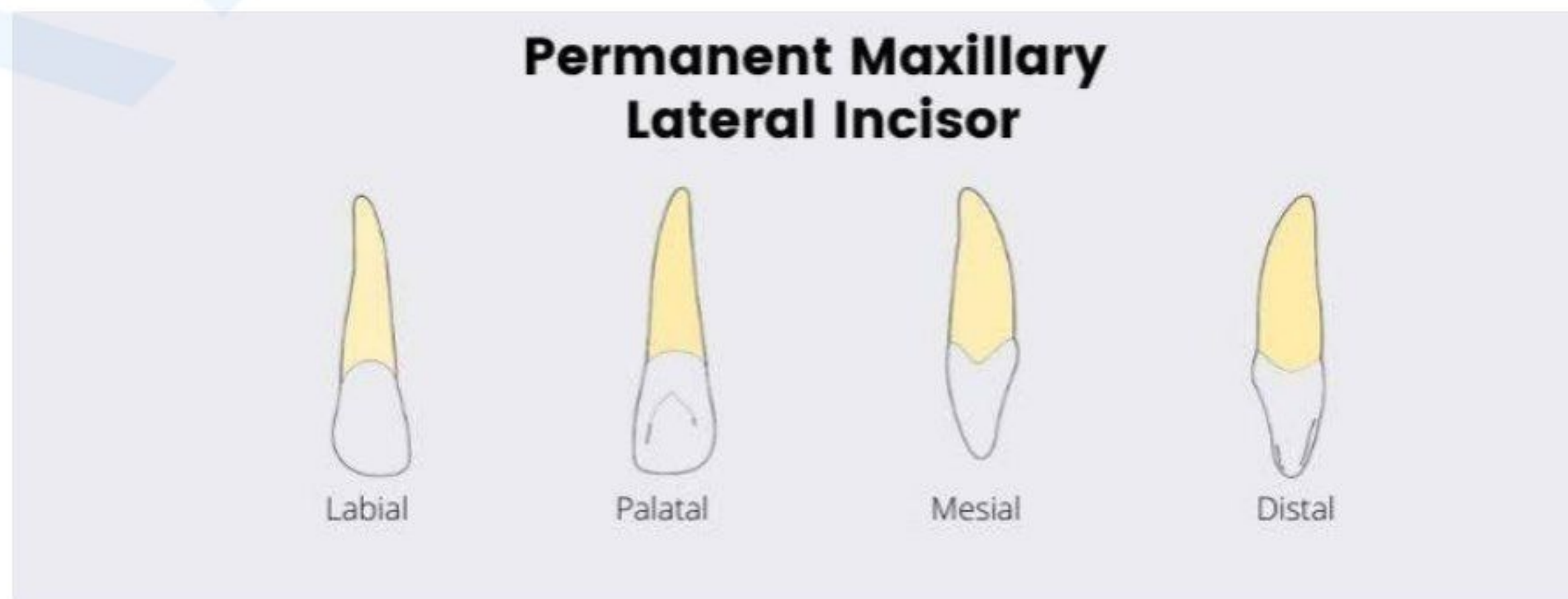
Root

- **Conical, straight, and round** on cross-section.
- Single root with labial, mesiolingual, and distolingual surfaces.
- Gradually tapers towards the **apex**, which is inclined distally.

Pulp

- **Spoon-shaped pulp chamber** with two cornua directed towards mesial and distal incisal angles.
- **Single root canal**.

Q2. Measurement and chronology of permanent maxillary lateral incisor



Permanent Maxillary Lateral Incisors

General Features

- Two in number, situated between the **central incisor** and the **canine** in the maxilla.
- Second tooth from the midline.
- Occludes with the lower **lateral incisor** and **canine**.
- Resembles the crown form of the maxillary permanent central incisor but is **shorter** and **narrower**.

Labial Surface

- May be **lobulated** and slightly convex.
- **Distoincisal angle** is more rounded than the mesioincisal angle.
- Incisal edge **slopes mesiodistally**.

Lingual Surface

- More convex, with height of contour in the **middle third** of the crown.
- Cervical line is **concave rootwards**, more so on the **mesial aspect**.
- The incisal edge is **thin**.

Mesial and Distal Surfaces

- Wider mesiodistally than labiolingually.
- **Wedge-shaped crown**.
- Mesial contour is convex, with height of contour at the junction of the middle and incisal thirds.
- The mesial surface area is **greater** than the distal surface area.

Incisal Edge

- May have **mamelons** and is placed in the labiolingual midpoint.
- Features a **thin incisal edge** with a more rounded distal edge.

Root

- Slightly longer in proportion to the crown compared to the dimensions of the central incisor.
- The apex is **distally inclined**.
- Developmental **depressions** present on the mesial and distal surfaces of the root.
- Root is more **ovoid** than the central incisor.

Pulp

- Contains a **small pulp chamber** with two cornua and a single root canal.

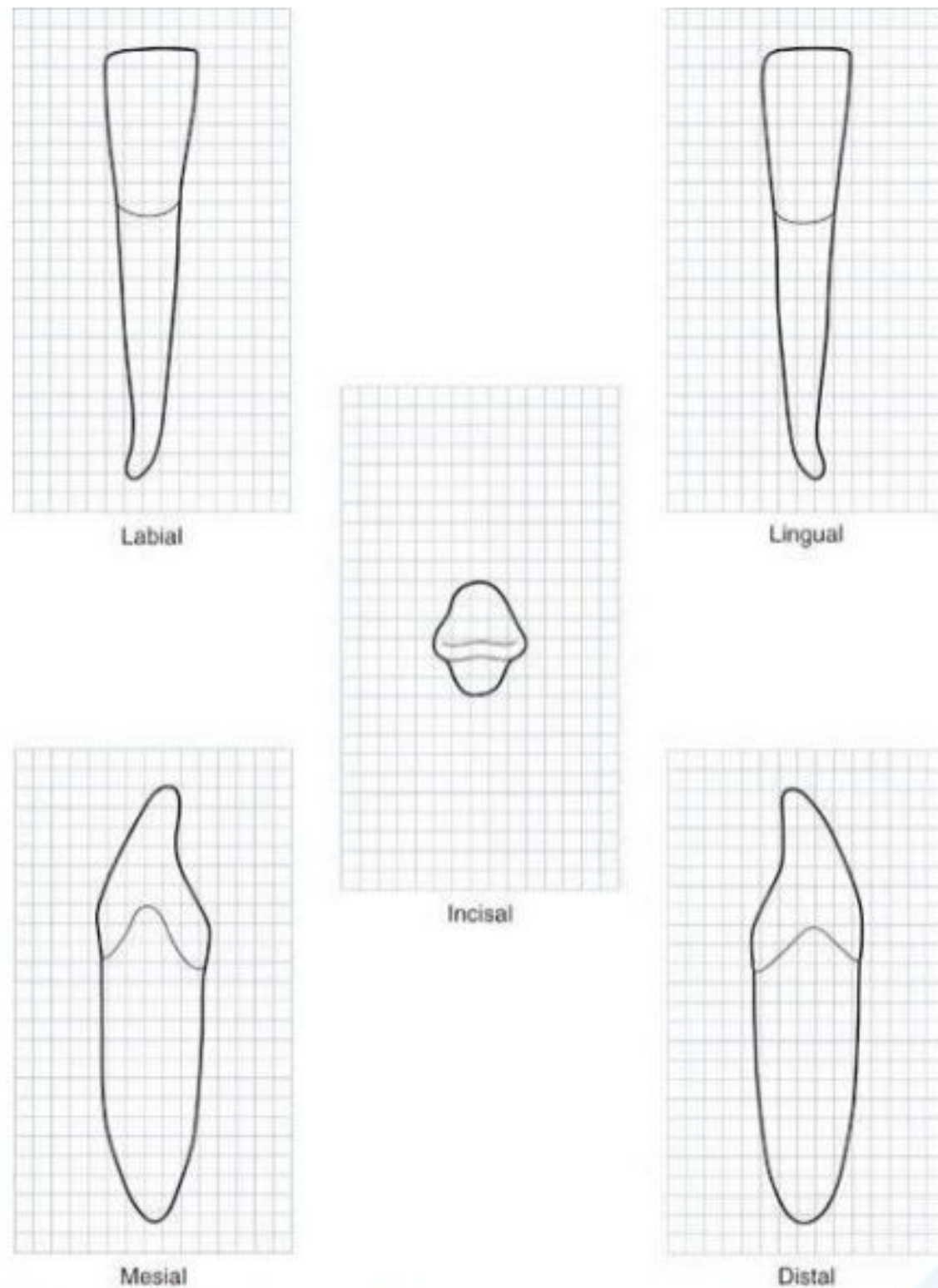
Q3. Difference between maxillary Central and lateral incisor.

Maxillary Central and Lateral Incisors

Feature	Maxillary Central Incisor	Maxillary Lateral Incisor
Number	Two (one on each side of midline)	Two (one on each side of midline)
Position	First tooth from the midline	Second tooth from the midline
Size	Larger than lateral incisor	Smaller than central incisor
Crown Shape	Broad, flat labial surface	More narrow and lobulated
Incisal Edge	Straight or slightly curved	Slopes mesiodistally
Mesioincisal Angle	Sharply defined	More rounded
Distoincisal Angle	Sharper	More rounded
Cervical Line	Convex rootwards	Concave rootwards
Labial Surface	Convex , prominent with two faint grooves	May be lobulated and slightly convex
Lingual Surface	Less prominent cingulum, smoother	More concave with a prominent cingulum
Lingual Fossa	Less pronounced	More distinct with a foramen caecum incisivum
Root Length	Shorter relative to crown	Longer relative to crown
Root Shape	Conical, straight	More ovoid and tapered towards apex
Developmental Features	Two mamelons prominent	May have mamelons , thinner incisal edge
Pulp Chamber	Larger pulp chamber with two cornua	Smaller pulp chamber with two cornua

Occlusion	Occludes with lower central and lateral incisors	Occludes with lower lateral incisor and canine
Overall Function	Primary function in cutting food	Assists in cutting food and aesthetics

Q4. Measurement and chronology of permanent mandibular Central incisor.



Mandibular Permanent Central Incisor

General Features

- Two in number, located on either side of the midline
- **Smallest** of all permanent teeth
- **Long but narrow**; crown height > width
- Occludes only with maxillary central incisor
- **Bilaterally symmetric**, forming almost a **90° angle** with the incisal edge

Labial Surface

- Convex but generally **flat** in the incisal half
- Tapers **evenly incisocervically** on mesial and distal borders

Mesial and Distal Surfaces

- Profiles mirror one another; **straight**
- Height of contour at **incisal edge**

- Wedge-shaped crown surfaces
- Inclines more **lingually** than maxillary incisors
- Cervical line is **concave rootwards**, more on mesial aspect
- **Straight incisal border** with sharp mesial and distal incisal angles

Incisal Edge

- **Straight** with three mamelons in newly erupted teeth
- Incisal edge is **lingual** to the labiolingual midpoint
- Edge is **perpendicular** to a line dividing the tooth labiolingually

Lingual Surface

- Very **indistinct lingual anatomy**
- Overall surface area is **less** than labial surface
- Concave except at cervical third, where it is **convex** due to a less-developed cingulum
- Marginal ridges are **not well-developed**; shallow lingual fossa
- Cervical line is **convex rootwards**

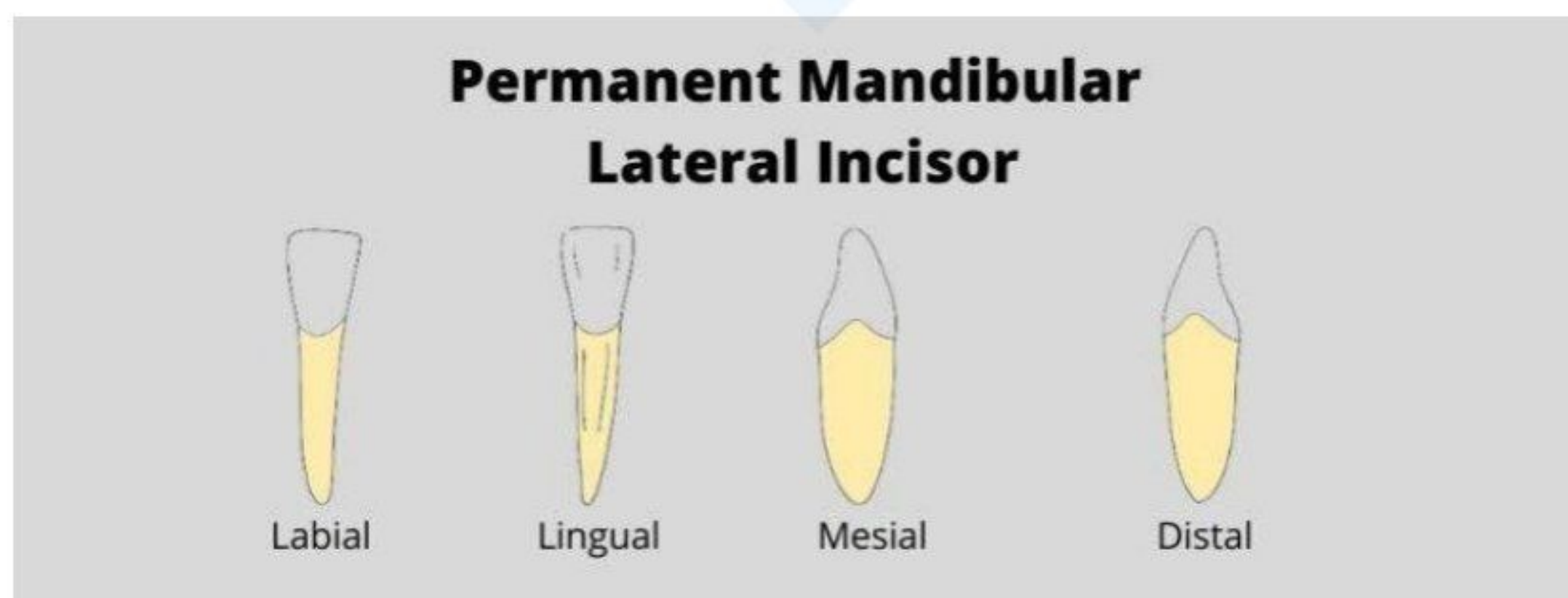
Root

- **Single-rooted**, flattened mesiodistally
- Distal groove is more pronounced than mesial groove
- Blunt root apex is **distally inclined**
- Ovoid in cross-section, **wider labiolingually** than mesiodistally

Pulp

- Spoon-shaped pulp chamber with two cornua directed towards mesial and distal angles of the incisal edge
- **Narrow root canal**; rarely has two root canals (buccal and lingual)

Q5. Measurement and chronology of permanent mandible lateral incisor.



Permanent Mandibular Lateral Incisor

General Features

- Two in number, second tooth from the midline
- Situated between the central incisor and canine
- Occludes with maxillary central and lateral incisors
- Resembles mandibular permanent central incisor but has key differences

Labial Surface

- Slightly **larger** than the central incisor
- Mesial contour is **straight**; height of contour at incisal edge
- **Wider** incisal edge compared to central incisor

Lingual Surface

- **Indistinct** lingual anatomy
- Marginal ridges are **better developed** than in the central incisor

Mesial and Distal Surfaces

- Mesial and distal surfaces are more pronounced
- Distoincisor angle is **more rounded** than sharp mesioincisor angle

Incisal Edge

- Incisal edge is **lingual** to the labiolingual midpoint
- Edge is not at a **right angle** to a line bisecting the crown
- Follows the arch curvature, making the **distal end** more lingually placed
- Incisal edge is **curved distally** and not straight

Root

- Root length is **greater** compared to the central incisor

Q6. Difference between mandibular permanent Central and lateral incisor.

Mandibular Permanent Central and Lateral Incisors

Feature	Mandibular Central Incisor	Mandibular Lateral Incisor
Number	Two (on either side of midline)	Two (second tooth from midline)
Size	Smallest of all permanent teeth	Slightly larger than the central incisor

Crown Shape	Long and narrow	Similar shape but wider incisal edge
Symmetry	Bilaterally symmetric	Less symmetrical compared to central incisor
Incisal Edge	Straight with three mamelons in new teeth	Curved distally, not straight
Labial Surface	Convex, generally flat in incisal half	Slightly convex, more lobulated
Mesial Surface	Straight with height of contour at incisal edge	Straight with a height of contour at incisal edge
Distal Surface	Straight, forms a right angle with incisal edge	More convex with rounded distofacial angle
Lingual Surface	Indistinct anatomy, concave	Indistinct anatomy, more convex with cingulum
Root Shape	Flattened mesiodistally	Slightly longer, more ovoid in cross-section
Pulp Chamber	Spoon-shaped with two cornua	Smaller pulp chamber with two cornua

Q7. Line angles and point angles in maxillary Central incisor.

Line Angles of the Maxillary Central Incisor

- **Mesiolingual:** Angle between mesial and lingual surfaces
- **Distolingual:** Angle between distal and lingual surfaces
- **Mesiolabial:** Angle between mesial and labial surfaces
- **Linguoincisal:** Angle between lingual and incisal surfaces
- **Labioincisal:** Angle between labial and incisal surfaces

Point Angles of the Maxillary Central Incisor

- **Mesiolabioincisal:** Angle formed by mesial, labial, and incisal surfaces
- **Distolabioincisal:** Angle formed by distal, labial, and incisal surfaces
- **Mesiolinguoincisal:** Angle formed by mesial, lingual, and incisal surfaces
- **Distolinguoincisal:** Angle formed by distal, lingual, and incisal surfaces